

AC9M3N06 • Year 3 Mathematics

# Mathematical Modelling in Practical Problems

Mathematical modelling turns a practical situation into a diagram, table or number sentence, then connects the solution back to the context.

## We are learning to...

use mathematical modelling to solve practical problems involving additive and multiplicative situations including financial contexts; formulate problems using number sentences and choose calculation strategies, using digital tools where appropriate; interpret and communicate solutions in terms of the situation

Understand  
contextChoose  
representation

Formulate

Solve

Check

Interpret

## Success criteria

- I can represent situations.
- I can formulate number sentences.
- I can use multiple operations.

## Key vocabulary

mathematical model

strategy

interpret

## Model a multi-step shopping problem

**3 × notebook**  
3 × \$4

**5 × pencil**  
5 × \$2

**Total \$22**

Three \$4 notebooks cost \$12 and five \$2 pencils cost \$10; the total is \$22.

## Choose operations from the situation



Some situations require more than one operation.

## Show what you know

### Model 4 packs of 6 shared among 3 children.

Think silently, show your reasoning, then explain it to a partner.

## Expected response

Three notebooks at \$4 each cost \$12.

### If students are unsure

Choose numbers and operations that match the question.

mathematical model

strategy

interpret